

# The Flavor and Charge Sectors — Preliminary Results: what the implementations and tests established

Tessera — cobordism subsystem (epic #410: #412, #413, #415, #417; #411/#416/#418 in flight)

## Abstract

This is a status report on the *Flavor and Charge Sectors* exploration track (epic #410), reporting the results actually *measured* as its tickets were built and tested. It is the companion to the plan note *The Flavor and Charge Sectors* (#420), which laid out the program; here we record what landed. Four results are merged and verified. (1) **Orientation is now geometric** (#412): the emergent charge read-out was made relabeling-invariant by applying the surface’s induced orientation symmetrically with the inputs, turning the color charge into the geometry’s Stokes invariant ( $\sigma_R = 0$  for a neutral input, exact; 3 for a colored one). (2) **The proton singlet is now exact** (#413): replacing the asymmetric prism extrusion with a *symmetric apex interior* drives the intertwining residual from  $4.26 \times 10^{-2}$  to  $4.5 \times 10^{-14}$  and the color-singlet overlap from 0.999 to 1.000000 — the transport is exactly equivariant — and it is now the default construction. A side finding: the spontaneous “photon” (a worldline relaxing through null) does *not* survive the symmetrization; it was triangulation-assisted. (3) The **Dirac–Kähler operator** (#415) is assembled and verified:  $(d + \delta)^2 = L$ , the Clifford relations exactly, the conserved current  $j^0$ , and the four-fold taste multiplicity. (4) The **electromagnetic field strength splits** into electric and magnetic parts by causal type (#417). Three further tickets (the Gauss-law charge #411, the bipartite revisit #416, the  $S^3$  spike #418) are in flight; the flavor synthesis (#414) is pending. All numbers are measured through the production C++ and its Python test suite; nothing here is projected.

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# 1 The program, and what this reports

The track finds the proton’s electroweak content (charge, flavor, spin/current) by splitting the behavior of the *existing* register — spatio-temporally and across form-degree — and by removing geometric arbitrariness, rather than bolting on parallel registers. The plan is documented separately (#420). This note reports the *measured* outcomes of the first wave of implementations. Each result below was landed as a pull request with its own tests, merged to `main`, and reproduced through the production `TransportCobordism / EigenstateSynthesis / DiracKahler` path; the figures are from those tests.

## 2 Geometric foundations

### 2.1 Orientation is geometric, not a vertex-label sort (#412)

The color charge of a window is a sum of per-hole periods. The inputs were already read in the surface’s induced orientation (the `endSignCovector`, a globally coherent  $\pm 1$  propagated across shared facets), but the *emergent result* was summed in each hole’s own sorted-vertex orientation — a function of the arbitrary vertex numbering. A relabeling could flip an individual hole’s term with no conjugating partner, scrambling the sum. The fix applies the result block’s induced signs symmetrically with the inputs. The effect is decisive: the signed carried charge becomes the geometry’s Stokes invariant.

| input                                       | signed $ \sigma_R $ (induced orientation) | raw $ \sigma_R $ (sorted) |
|---------------------------------------------|-------------------------------------------|---------------------------|
| neutral ( $q\bar{q}$ triple, $\Sigma = 0$ ) | 0.0000 (exact)                            | 0.2258                    |
| colored ( $R, G, B$ , $\Sigma = 1$ each)    | 3.0000 (exact)                            | 1.0219                    |

The signed sum is exactly the Stokes charge  $\sigma_R = -(\sigma_A + \sigma_B + \sigma_C)$ ; the raw sum is neither, and is label-dependent. The fix is verified by a relabeling-invariance test (the track’s  $G_6$  invariant): under seeded permutations of the base-surface vertex ids the signed  $\sigma_R$ , singlet overlap, and intertwining residual are invariant up to a single global sign, while the raw sum drifts. The companion clarification — that the read-out’s two notions of “orientation” are distinct — led to renaming the Causal-Dynamical-Triangulations ( $t_i, t_f$ ) class to `TemporalOrientation`, reserving the *spatial* induced orientation for `ChainComplex::endSignCovector`.

### 2.2 The symmetric apex interior: the proton singlet is now exact (#413)

The trivalent junction  $W_{ABC}$  was built by extruding the holed base surface  $\times I$  and tetrahedralizing each triangular prism by the Freudenthal “staircase” — whose diagonal is chosen by a vertex-label sort. That choice is not a symmetry of the geometry, and it — not the metric — was the documented source of the intertwining residual  $\|MP_{\text{in}} - P_{\text{out}}M\| / \|M\| \approx 4.26 \times 10^{-2}$  that held the color singlet at 0.999.

The replacement is a *symmetric apex stacking*, with *no* vertex-label sort anywhere. Each base triangle cones up to a single *face-apex* vertex (a (3,1) tetrahedron) and down from the apex to the top copy (a (1,3) tetrahedron); the gap between two adjacent apex-columns over a shared base edge is an octahedron split along the *canonical dual edge*  $f_1 f_2$  (the two face-centres) into four tetrahedra. Because the dual edge is fixed by the two incident faces rather than by ids, the subdivision is equivariant under the icosahedral symmetry. The transport then intertwines the color  $\mathbb{Z}_3$  *exactly*:

|                                                                      | prism (Freudenthal)   | symmetric apex (now default) |
|----------------------------------------------------------------------|-----------------------|------------------------------|
| intertwining residual $\ MP_{\text{in}} - P_{\text{out}}M\  / \ M\ $ | $4.26 \times 10^{-2}$ | $4.5 \times 10^{-14}$        |
| color-singlet overlap                                                | 0.999                 | 1.000000                     |
| valid manifold ( <code>dualComplexValid</code> ) / $b_1$             | yes / 11              | yes / 11                     |

The construction is exposed as `Spacetime::symmetricStackCells` and is the default interior on `TripartiteRegisterTopology` (`set_symmetric_interior(false)` selects the legacy prism). The twenty existing junction tests pass unchanged on the symmetric default — no re-baselining was needed, the residual simply moved inside their tolerances — so the color-confinement skeleton (charge conservation exact at  $\sim 10^{-14}$ ,  $\sigma \rightarrow 0$ , the  $\mathbb{Z}_3$  spectrum  $\{1, \omega, \omega^2\}$ ) is preserved while the singlet becomes exact.

**Remark 1** (The spontaneous photon was triangulation-assisted). *On the prism, making the cross-layer worldlines timelike and relaxing drives one worldline through null ( $\ell^2 \rightarrow 0$ ) — the predicted “photon.” On the symmetric interior the same seed does not produce it: through 120 relaxation iterations all 450 worldline edges stay uniformly timelike at  $\ell^2 = -0.3$ , with zero near-null edges. The symmetric relaxation preserves the worldline symmetry; the prism’s photon was triggered by the triangulation asymmetry singling out one edge. The physical reading is that a photon should emerge from a genuine symmetry-breaking source, not from the bare uniform seed — the symmetric interior is the more honest setting, and a source-driven photon is the natural follow-up. (The Lorentzian seed itself works on both interiors: a structural time rule,  $\text{bottom} = 0$ ,  $\text{apex} = 1$ ,  $\text{top} = 2$ , makes the  $\text{surface} \leftrightarrow \text{apex}$  and  $\text{bottom} \leftrightarrow \text{top}$  edges the timelike worldlines, and it stays equivariant.)*

### 3 The spin / current sector: Dirac–Kähler (#415)

The inhomogeneous cochain  $\Omega^\bullet = \bigoplus_k \Omega^k$  is the discrete Dirac–Kähler field, and  $D = d + \delta$  is its Dirac operator. Assembled from the integer boundary maps and metric weights (reusing the convention the Hodge Laplacian already squares), the construction reproduces the expected algebra to machine precision:

| property                              | measured                          | meaning                                                        |
|---------------------------------------|-----------------------------------|----------------------------------------------------------------|
| $(d + \delta)^2 = L$                  | residual $\sim 6 \times 10^{-15}$ | $D$ is a square root of the Hodge Laplacian                    |
| $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$ | residual 0.0 (exact)              | the Clifford algebra of the 1-cochain action                   |
| $j^0 = \bar{\Phi}\gamma^0\Phi$        | $= \langle \Phi, \Phi \rangle_W$  | the conserved (Noether) charge density                         |
| multiplicity                          | $= 4$                             | the taste degeneracy $\Lambda(\mathbb{C}^4) = 16 = 4 \times 4$ |

The gamma generators are exposed as the  $16 \times 16$  Clifford action of unit 1-cochains on the form fiber  $\Lambda(\mathbb{R}^4)$  — the literal realization of “gammas = the Clifford action of edges.” The four-fold multiplicity (the lattice Dirac–Kähler “doubling”) is the candidate flavor/taste index. The alternative representation — the irreducible  $4 \times 4$  Dirac matrices  $\gamma^\mu$  — is recorded as a documented extension point: it is the image of the same generators under the Chevalley/Kähler–Atiyah isomorphism  $\Lambda(\mathbb{C}^4) \rightarrow M_4(\mathbb{C})$ , a sibling accessor that would change only the gamma dimension ( $16 \rightarrow 4$ ) and make the multiplicity an explicit taste degeneracy;  $D$  and  $j^0$  are representation-independent. The current’s time component  $j^0$  is the charge density; its cross-check against the gauge-field Gauss-law charge is the subject of §4 and is deferred to #411.

## 4 The charge sector

### 4.1 Electric and magnetic fields by causal type (#417)

The electromagnetic field strength is a 2-cochain  $F = dA \in \Omega^2$ . On the Lorentzian complex it splits by the causal type of the plaquette: a plaquette is *electric* if any of its edges is timelike (a temporal leg), and *magnetic* if it is purely spacelike. This is implemented as `EigenstateSynthesis::curvatureFromConnect` ( $F = dA$ , the same induced-orientation signed edge sum the periods use) and `fieldStrengthSplit`, returning the electric and magnetic parts and their disjoint, complete cell index lists. The tests confirm the split tracks the causal structure:

- on a purely spacelike (static) seed:  $\|E\| = 0$  exactly,  $B = F$  on every cell;
- under timelike flux:  $\|E\| > 0$ , and the electric cells equal the independently-recomputed timelike-leg plaquettes;
- $E + B = F$  to  $10^{-12}$ , and  $F = dA$  is gauge invariant ( $A \rightarrow A + d\chi$  leaves  $E, B$  unchanged).

The reader is purely observational: the proton sector is bit-for-bit unchanged with it present or absent. Because  $E/B$  is read off the cochain by causal type, it is *interior-shape-independent* — it holds equally on the prism and the symmetric apex interior.

### 4.2 Electric charge as a Gauss-law holonomy (#411, in flight)

The charge ticket reads electric charge as the temporal-sector Gauss law,  $Q = \oint \star F$ , a gauged- $U(1)$  holonomy of the field of §4.1 — expected to be metric-robust and quantized, unlike a hand-weighted color covector — and to verdict the fractional- $(\frac{1}{3})$  charge link to triality. A required cross-check, handed off from #415, is that this Gauss-law  $Q$  agrees with the Dirac–Kähler current’s conserved charge  $j^0$  of §3. This work is in progress; results will be folded in.

## 5 The flavor and binding sectors (in flight)

Three further pieces are under construction and are reported here only as scope:

- **Flavor without a register (#414, pending).** Seeks the u/d label (proton  $uud = +1$  vs neutron  $udd = 0$ ) in the spatio-temporal split or the Dirac–Kähler taste multiplicity of §3, not a parallel hole register. It depends on the Gauss-law charge (#411).
- **The bipartite revisit (#416, in flight).** Tests whether a twisted (orientation-reversing) bipartite tube projects onto the antisymmetric diquark  $\bar{\mathbf{3}}$ , and whether a connected two-vertex path reaches the singlet — on the *bipartite/prism* path, which the symmetric apex interior (a tripartite-junction construction) does not replace.
- **The  $S^3$  dimensional spike (#418, in flight).** Scopes whether the apex/dual-edge octahedron split generalizes to a triangulated  $S^3$  base. The dimensional question is *not* settled by reducing to a lower-dimensional sector: the program is fully emergent, and dropping to 2+1 D (an  $S^2$  slice) would change the physics rather than answer it. The honest route is to *generalize* the apex interior dimension-generically. The interior connectivity follows from the base layer’s coface/dual structure: cone each top  $d$ -simplex to an apex, and reflect apex-connectivity across a  $(d-1)$ -facet shared by two cofaces (with lower faces shared by  $k$  cofaces mirrored analogously). In 2D this is exactly the dual-edge octahedron split; in 3D it cones tetrahedra to cell-apexes across shared triangles. The current `Spacetime::symmetricStackCells` triangle-only special case is an implementation gap, not a wall. This coface-mirroring generalization,  $S^3$  over a tetrahedral base, and the measurement it enables are carried by #429 (see also the assessment below).

## 6 Faithfulness and no-regression

Every result above was landed under a fixed set of proton invariants (the track’s  $G_1$ – $G_8$ ): the singlet overlap, confinement  $\sigma \rightarrow 0$ , the exact Stokes charge conservation,  $b_1 = 11$  with a valid manifold, the color  $\mathbb{Z}_3$  spectrum, relabeling- invariance, determinism, and emergent-first read-out. They held throughout: the geometry change (#413) preserved the color sector exactly while sharpening the singlet, the orientation change (#412) preserved the manifold and the spectrum while making the charge a label-independent invariant, and the Dirac–Kähler and  $E/B$  additions (#415, #417) are observable-only — the proton observables are unchanged with them present. No golden result drifted; the twenty junction tests and the per-ticket suites are green on `main`.

## 7 Assessment and open questions

**What is established.** The geometry is now honest (no label-sort artifacts), the proton color singlet is *exact* (overlap 1.000000, residual  $10^{-14}$ ), the charge read-out is the geometry’s Stokes invariant, the field strength carries a faithful  $E/B$  split, and the Dirac–Kähler operator gives genuine  $4 \times 4$ -sized Clifford structure with a conserved current and a four-fold taste index — all verified to machine precision and merged.

**What is open.** The electric charge as a protected Gauss-law holonomy and its agreement with  $j^0$  (#411); whether flavor is the taste multiplicity or the spatio-temporal split (#414); whether two bipartite steps reach the singlet (#416); and the higher-dimensional electroweak content over  $S^3$  (#418). The dimension is answered not by a reduced sector but by generalizing the apex interior via coface-mirroring (#429): build  $S^3$  over a tetrahedral base and read off what emerges. A forward direction prototyped there is an *emergent twist*: each apex is a centre of point-reflection (its neighbour edges inverted through it to form the next slice, plus a cap), and since composing reflections is a rotation, iterating reflect-and-cap over many slices may accumulate a net screw/monodromy along time. Such a twist is read off the relaxed geometry (unlike the hand-supplied prism twist) and makes  $W$  a mapping torus (a twisted product); a  $\mathbb{Z}_3$  twist would be the colour singlet’s  $\omega$ -phases, a  $2\pi \rightarrow -1$  twist the spinor sign (spin), and a  $U(1)$  twist a gauge holonomy (charge). Two deeper questions persist from the plan: the cross-sector *total antisymmetrization* (the  $SU(6)$  **56** that makes three labels a proton, not just color  $\times$  flavor  $\times$  spin), and whether isospin is realized as a genuine  $SU(2)$  with a ladder rather than a static u/d label. Finally, Remark 1 reframes the photon: it should be *sourced*, not spontaneous, on the symmetric interior — a concrete next experiment.

**Verdict.** The color-confinement skeleton is now exact and the surrounding machinery (orientation,  $E/B$ , the Dirac–Kähler current) is in place and verified; the electroweak read-outs that consume it are the work in flight. The track is on its planned path, with every landed step measured rather than asserted.